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Pollination of *Anemopaegma album* (Bignoniaceae) with focus on floral nectar as the mediator of interactions with mutualistic and antagonistic bees

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ABSTRACT

We focused our study on the pollination of *Anemopaegma album*, a Bignoniaceae species from a seasonally dry habitat, and on the features of primary attractive, the nectar, which mediate the interaction with mutualists and antagonists bees. We investigated the floral biology and identified pollinators and nectar robbers through field observations; we determined the mating system by performing controlled experiments; we described the floral structure with emphasis on the arrangement of reproductive structures and on the floral nectary and nectar chamber using scanning electron microscope for surface micromorphology and optical microscope for nectary histology; we monitored nectar production during the first day of anthesis and the weather conditions around plants. We registered medium and large-sized bees visiting *A. album* flowers throughout the daylight period acting as pollinators (Euglossini and Bombini) and as nectar robbers (*Oxaea* and *Xylocopa*). Plants from the focal population showed obligate outcrossing depending on bees for sexual reproduction; and flowers showed herkogamy mechanisms that favour outcrossing. The nectar chamber is formed by the constriction caused by the dilated basis of filets covered densely by trichomes and the nectary showed histological features related to the production of high amount of nectar in a short period. In fact, the highest volume of nectar is available when flowers open; after this, just small quantities are added along the day. The hottest and driest period was around midday, when just the nectar robber bee from genus *Oxaea* visited the flowers. Bees that promote *A. album* outcrossing used diluted nectar as a floral resource and concentrated their visits during the morning when the highest volume of nectar was available and the weather conditions were more suitable.

Keywords: flower-bee interactions, seasonally-dry habitats, floral nectary, hercogamy, mating system.

INTRODUCTION

Plant-pollinator mutualism provides an ideal scenario to investigate the interplay of ecological and evolutionary effects (Mitchell *et al.* 2009), allowing an understanding of biological adaptations at all levels (Willmer 2011). So, information on the biotic and abiotic pressures on floral primary attractives, especially on nectar that is the most common floral resource to pollinators, could be essential to understand the outcome of these connections to plant reproduction (Herrera 2000, Sun *et al*. 2013) and to plant evolution (Johnston 1991, Galen 1996).

Concerning the biotic pressures on nectar many plants interact, besides pollinators, with nectar cheaters, consequently participating in the energy supply of different functional groups. Nectar robbing (*sensu* Inouye 1980) is a common larceny in Bignoniaceae (Milet-Pinheiro & Schlindwein 2009), and the effects of nectar robbers can be positive or negative, depending on several characteristics of the plant species involved in the interaction (Maloof 2001). So, both mutualistic and antagonistic interactions should be considered in ecological studies of pollination and floral biology (Burkle *et al*. 2007).

Moreover, nectar is an aqueous solution and its production depends largely on the water availability at the site during the flowering period (Petanidou 2007). Variations in the climatic conditions under which plants, pollinators and robbers interact can influence nectar features in different ways (Kenoyer 1916, Corbet 1990, Silva & Dean 2004), and they can be especially important in dry areas (Nicolson 1995). These climatic influences can have consequences for pollinator needs and behavior, ensuring the maintenance of pollinator interactions through the offer of a suitable resource throughout anthesis. In fact, in zones with dry atmospheres where evaporation is intense, ecophysiological processes can reduce the variability in nectar concentration throughout the day, which may ensure a more stable and predictable source of energy for pollinators (Nicolson & Nepi 2005). In addition, nectar may constitute the main and in some cases the only form of water intake for floral visitors (Petanidou 2007). However, basic information on floral biology and floral nectar exploiters in plants from drier ecosystems is poorly explored to date.

Here, we choose the plant species *Anemopaegma album* (Bignoniaceae) as a model to study the plant-bee interaction mediated by floral nectar in a typically seasonally dry habitat in Brazil. The general aims of our study were to verify the plant-pollinator dependence for plant sexual reproduction, and the pollination ecology of *A. album* including the study of nectar exploitation by two distinct functional groups of visitor: legitimate visitors (pollinators) and non-legitimate visitors (nectar robbers). We focused on study on these four questions : (1) What is the pattern of development of male and female phases during anthesis? (2) What is the bee behavior on *A. album* flowers: pollinators or nectar robbers? (3) Does *A. album* depend completely on pollen vector for sexual reproduction? (4) What's the relationship between floral visits, weather conditions, flower nectary and floral nectar in this seasonally dry ecosystem?

METHODS

Study area and plant species—The experiments and field observations were performed at the 'State Park of Grão Mogol' in Minas Gerais state, Brazil in a chain of mountains named the 'Cadeia do Espinhaço', that represents a contact zone between the 'cerrado' and 'caatinga', meaning savanna and steppe savanna vegetation types *sensu* Walter (1986), respectively. The vegetation is xeromorphic (Giulietti *et al.* 1997) and dominated by plants with a high fixation capacity for the substrate, and the plants are desiccation-tolerant or resistant to water stress (Rapini et al., 2008). The climate is seasonally semi-arid, with a dry season occurring between April and September and a mean annual rainfall of approximately 1182 mm that is distributed over the other 6 months, when 80% of the total annual precipitation falls. The mean temperatures vary from 13.3° to 30.6° C during the dry season and from 18.7° to 31.8° C during the rainy season (Ramos *et al.* 2009). *Anemopaegma album* study population was found in rocky mountain vegetation at 'Cadeia do Espinhaço' at 900 to 1400 m in altitude.

The genus *Anemopaegma* (Bignoniaceae) is pollinated by large- and medium-sized solitary bees (Gentry 1974) and although the most of the species occur in forests, we can find some of them, as *Anemopaegma album*, growing in seasonally dry areas (Lohmann 2003, 2011). All experiments and field observations were performed in the summer, January 2010, during the population blooming.

Flowers events during anthesis and floral visitors—To identify the nature of interactions with each floral visitor we recorded the anthesis events in 10 plants with an emphasis on floral morphology, opening time, longevity, period of pollen availability using acetocarmine and of stigma receptivity, which was inferred from hydrogen peroxide analysis (Dafni *et al*.

2005). Additionally, we registered floral traits related to primary and secondary attractiveness to floral visitors as the presence of nectar, the color of the floral elements and the indicative of osmophores presence as evaluated using neutral red (Vogel 1990). We monitored the flowers to check for visitors at different times of day throughout five days during the flowering period for a total of 40 hours of observation from dawn (06:00 h) until night (22:30 h). Visitor behavior was described on the basis of field observations and photographic analyses. We recorded the time and duration of visits in addition to the identification of body regions that came into contact with the anther/stigma, and the type of resources collected by floral visitors. We calculated the mean frequency of visits per daylight hour and per day using days as replicates.

Floral structure and nectary characterization—We dissected functional flowers in order to describe its structure, especially the basis of corolla tube where is located the floral nectary and nectar chamber, which is directly related to nectar production and storage, respectively. Dissected flowers were used for drawing and for micromorphology characterization through scanning electron microscopy (SEM). For SEM we fixed the base of corolla tube from functional flowers in 2.5% glutaraldehyde (0.1 M phosphate buffer, pH 7.2), dehydrated in an ethanol series, dried to critical point, and sputter-coated them with 10 nm of gold (Robards 1978).

We performed anatomical analyses of floral nectary disk using samples from functional flowers fixed in glutaraldehyde (2.5% with 0.1 M phosphate buffer, pH 7.3). The samples were left overnight at 4°C and post-fixed with 1% osmium tetroxide (OsO4) in the same buffer for 2 h at room temperature. Subsequently, they were dehydrated in a graded series of acetone solutions, and embedded in Araldite resin. The semi thin sections (0.5µm) were stained with 1% toluidine blue (pH 4.3) (O'Brien *et al.* 1964) and evaluated under light microscope.

Mating system—To verify the level of dependence on pollinators for reproduction, we performed experiments to verify the occurrence of self-compatibility in the study population. We used 29 flowers from 29 plants for manual deposition of pollen on stigma surface from the same flower. Additionally, we used 153 flowers from 52 plants for manual deposition of pollen mix on stigma surface to verify the potential seed set in experimental 'absence of pollinator limitation'. The pollen mix was prepared with donor (male) flowers from five to seven plants apart at least 20 meters from receptor (female) flowers. For both treatments, we isolated pre-anthesis buds from floral visitors using bridal veil bags. On the second day of anthesis, we pollinated the flowers using thin disposable sticks. We bagged the flowers again immediately following manual pollination and monitored them weekly until fruit development or flower abscission. Natural fruit set was also evaluated. For this measurement, we recorded the number of fruits formed from 1634 labeled flowers in 142 plants that were not protected against any type of visitors.

Microclimate—To characterize the microclimate where the *A. album* plants and floral visitors occurred, we recorded the temperature and relative humidity from 07:00 to 19:00 h every 2 hours using a portable thermo hygrometer.

Floral nectar—We monitored nectar volume and concentration. First we selected a set of pre-anthesis buds without any visible damage and protected them with bridal veil bags to prevent nectar withdraw by floral visitors during monitoring, according to Corbet (2003). Subsequently, we withdraw all the nectar from freshly opened flowers (at 0700 h on the first day of anthesis) (n_{0h} = 30 flowers) with graded glass syringes to measure the total volume of nectar per flower (µL) at the beginning of anthesis. We also measured the sugar concentration with a digital refractometer (%, weight /weight) *in situ*. After 12 hours, in the evening (at 1900 h on the first day of anthesis), we took a second set of flowers to measure the nectar volume and concentration ($N_{12\ h}$ = 18 flowers). Analyses of variance were used to test whether nectar secretion differed between the two sampled moments: 0700 and 1900 h, and statistics reported were based on untransformed data and were performed in R version 3.1.1 (R Core Team 2014).

RESULTS

The pollination ecology of *A. album*—We observed one to four flowers opening per inflorescence per day; approximately 20% opened from 0500 h to 0600 h, and approximately 80% opened from 0600 h to 0700 h. The anthesis lasted approximately 4 days (*n* = 111 flowers,

53 plants). We considered that a flower had opened when the corolla lobes were completely curved away. The recently opened flowers were yellow inside the tube and pale yellow to whitish outside and on the lobes (Fig. 1A). At the beginning of anthesis, the anthers were opened, pollen was already available, and the stigma had separated lobes and exhibited a light reaction to hydrogen peroxide. The stigma showed a strong reaction during the second day of anthesis, suggesting a peak of receptivity, and we did not observe any reaction on the third day, indicating that the duration of the female phase in *A. album* was approximately 2 days. We observed stigma closure a few seconds after touching it. A specific zone, which was composed of a line of dots distributed throughout the internal surface of the tube following the longitudinal grooves on the floor of the corolla, was stained with neutral red. Additionally, we recorded many red dots that were dispersed on the margin of the lobes and at the landing area. Nectar secretion started at the pre-anthesis stage, and some nectar was available in the basal portion of the corolla tube at the moment of flower opening.

Bees belonging to Apidae: Apinae (Michener 2000) visited the *A. album* flowers in a legitimate manner from approximately 0700 h until 1200 h and from 1600 until 1900 h. Medium-sized bees belonging to the Euglossine and Bombini tribe visited the *A. album* flowers acting as potential pollinators. We registered two visits of a large-sized bee, probably belonging to Centridini, but it was not possible to collect it for taxonomic identification. This bee also visited the flowers in a legitimate way, once around 10:00 h and the other time around 18:00 h. We considered legitimate floral visitors to be those that were able to perform pollination through contact with anthers that were filled with pollen during resource collection and with the receptive portion of the stigma. Additionally we verified that when a legitimate visitor left the tube carrying pollen on the backside of its thorax or abdomen, it only contacted the sterile portion of the stigma, favoring outcrossing.

The Euglossini bees visited *A. album* flowers in a legitimate way in the morning, from 0900 to 1200 h and in the afternoon from 1600 to 1900 h on four different days. These bees were observed visiting 1 to 2 flowers per hour in the morning and 3 flowers per hour in the afternoon. Additionally, these bees visited only one flower per plant in all the records. We registered a mean of 0,40 $\pm$ 0,37 flowers visited per hour during a daylight period by Euglossini bees. Additionally, these bees performed 0,22 $\pm$ 0,26 (mean $\pm$ sd) sequences of visits per hour. After a quick investigation, the bees entered the flower by moving towards the bottom of the tube, apparently searching for nectar, and they left the flower 3-10 seconds later.

The Bombini bees always visited *A. album* flowers in a legitimate way, and they concentrated their visits during the morning and evening, from 0700 until 1100 h, when we registered 1 to 8 visits per hour; in the afternoon they visited flowers from 1700 to 1900 h, when we registered 1 to 7 visits per hour. These bees foraged in one to four flowers per inflorescence, and they commonly visited each flower only once. We registered a mean of 0,82 + 0,59 of flowers visited per hour by Bombini bees and 0,20 $\pm$ 0,13 (mean $\pm$ sd) sequences of visits per hour. These bees flew around the inflorescences and hovered in front of each flower for a few seconds, subsequently introducing their heads into the flower and using the two lower lobes as landing support, after entering their whole body as deep as possible inside the floral tube (Fig. 1B). During the period in which these bees were inside the flower, which varied from a few seconds to more than one minute, they apparently performed nectar collection from the nectar chamber and contacted the backside of their bodies with the internal surface of the stigmatic lobes during their entrance, followed by contact with the anthers. Finally, the bees generally flew towards the flowers of other inflorescences on the same plant or to neighboring plants after the first visit recorded in the field.

The *A. album* flowers were also visited by nectar robbers and pollen thieves. Bees from the genera *Oxaea* and *Xylocopa* (Fig. 1C) acted as nectar robbers. Like the pollinators, *Xylocopa* bees used to visit the flowers in the morning and the evening, from 0700 to 0800 and from 1600 to 1900 h, performing one sequence of visits per hour. *Oxaea* visits occurred exclusively between

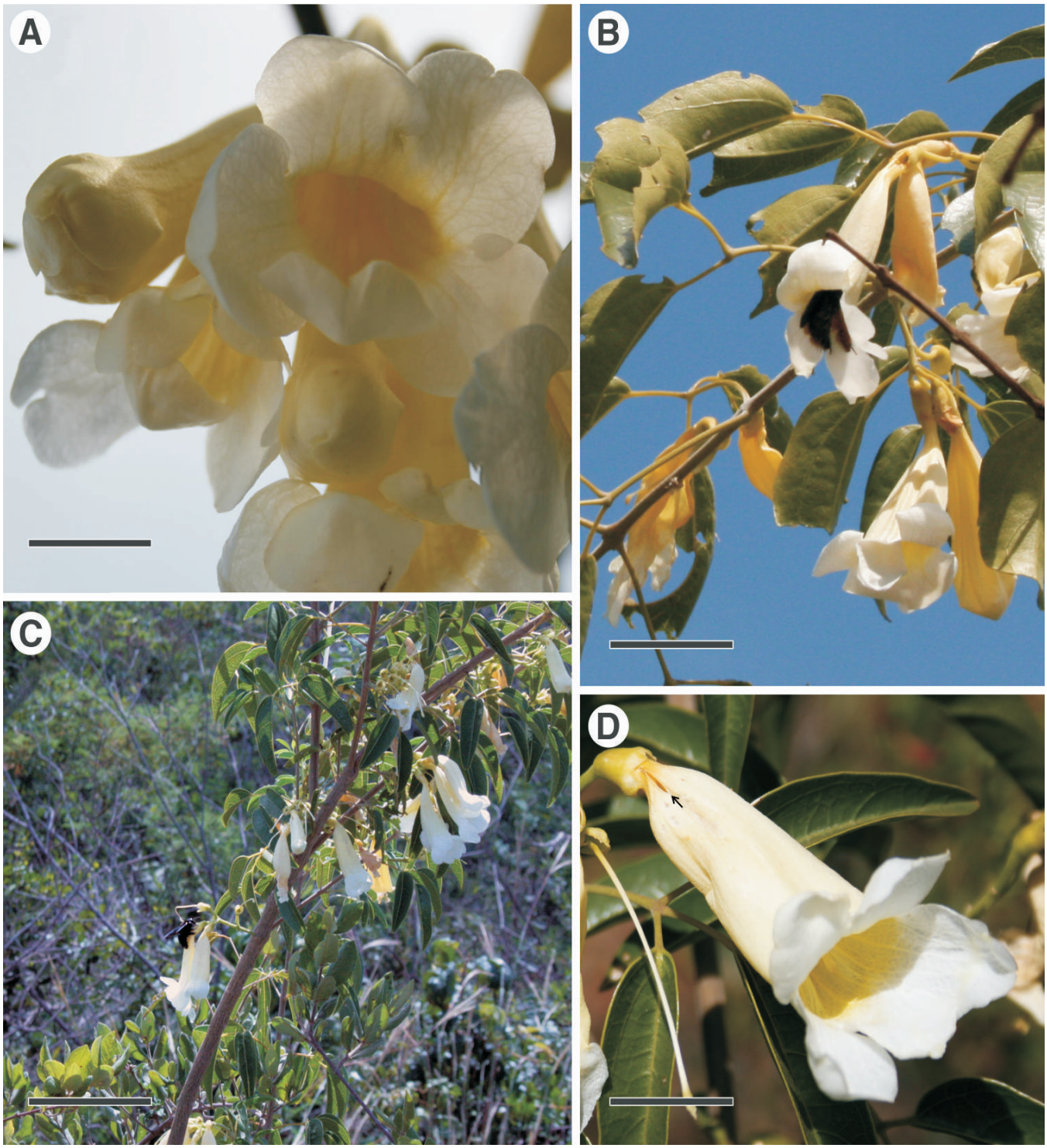


Fig. 1- Flower and floral visitors of *Anemopaegma album* (Bignoniaceae) at the "State Park of Grão Mogol", MG, Brazil. **A.** An inflorescence at approximately 7:00 h exhibiting a floral bud approximately 24 hours before anthesis, opening flowers and a completely opened flower. **B.** A pollinator bee entering the floral tube. **C.** A bee from the genus *Xylocopa* robbing nectar. **D.** A slit performed by nectar robber at the base of floral tube. Scale bar: (A) 1 cm, (B) 4 cm, (C) 7 cm, and (D) 2 cm.

0800 and 1400 h, being more concentrated around midday, when it performed until six sequences of visits per hour. These robber bees, from both genera, landed externally over the floral tube (Fig. 1C) and introduced their mouth apparatus between the calyx and corolla, forming a slit on the base of the corolla (Fig. 1D) from which they could withdraw the nectar stored in the chamber. These bees never contacted the reproductive structures of *A. album* flowers. We registered a mean of 1,7 $\pm$ 2,3 legitimate visits/ day and a mean of 1,9 $\pm$ 2,1 non-legitimate visits/ day of bees that exploit nectar as energetic resource in *A. album* flowers.

Additionally, we observed small bees collecting pollen actively from the anthers, from 0600 to 0800 h and around 1800 h, acting primarily as pollen thieves. These bees entered the flower without contacting the internal surface of the stigma and started to collect pollen, and they packaged the pollen on their third pairs of legs. During this procedure, they mostly contacted the external and, eventually, the internal surface of stigmatic lobes during some maneuvers, preferentially performing self-pollination.

Floral structure and nectary anatomy—The floral morphology and the arrangement of reproductive organs revealed herkogamy mechanisms (Fig. 2A). The anthers and stigmas were positioned so that the receptive surface of the stigma was contacted first when the visitor entered the tube in search of nectar, and the pollen was subsequently picked up on its back. When the visitor left the flower, the visitor only contacted the sterile external surface of the stigma. To access the resource stored in the nectar chamber (Fig. 2A) the floral visitors have to transpose the constriction created above the ovary by the enlarged base of epipetalous stamens densely covered by trichomes (Fig. 2B).

The nectary disk is placed in the lower portion of nectar chamber, below the ovary (Fig. 2C) and it consists of three well defined regions: (i) a uniseriate epidermis covered by a irregularly ticked cuticle, (ii) a nectary parenchyma composed of 3-5 layers of small and densely stained cells with compact arrangement and (iii) a subnectary parenchyma formed by 15-20 layers of loosely arranged cells supplied with phloem only (Fig. 2D). The subnectary parenchymal cells had larger packages of starch grains, which became smaller towards the epidermis (Fig. 2E).

Mating system—The population of *A. album* evaluated in this study showed obligate outcrossing with a total dependence on pollen vectors for sexual reproduction; no fruit set was registered in flowers that received self-pollen. We registered 2.4% of natural fruit set and 30.7% of fruit set in the hand-pollinated flowers with a mix of pollen, showing the prevalence of allogamy in the study population.

Microclimate—*Temperature and relative humidity variations during bee flower visits.* Bee visits to *A. album* flowers started when the air temperature was 22.2°C and the relative humidity (RH) was 67% (Fig. 3A, B). Legitimate visits occurred during two separate periods of the day as follows: (i) approximately 0700 until 1200 h and (ii) from 1600 until 1900 h (Fig. 3A). No legitimate visits were observed between 1200 and 1600 h, but we registered visits performed exclusively from the nectar robber, *Oxaea* sp, from 1200 until 1400 h (Fig. 3A), when the temperature reached 45.5°C and the relative humidity was 17% (Fig. 3B). At approximately 1700 h, when the temperature was lower 31.7°C and the RH was higher 33% (Fig. 3B), the legitimate visits of medium- and large-sized bees increased again (Fig. 3A).

The floral nectar production in *A. album*—The nectar secretion started at pre-anthesis, so when the flowers opened, around 0700 h, there was 30 µL (0-85) (median, max-min) of nectar accumulated inside the nectar chamber, which didn't change significantly its volume after 12 hours of anthesis, around 1900 h, presenting 35.5 µL (0-102) (median, max-min) of nectar accumulated inside the nectar chamber (Fig. 3C; $F_{(1,46)} = 1.33$; $p = 0.25$). The sugar concentration of opened flowers was 29.55%, w/w (19.40-34.8), (median, max-min) with little variation in the first 12 hours of anthesis showing values of 27.6 29.55%, w/w (11.80-34.3), (median, max-min) (Fig. 3D; $F_{(1,37)} = 3.44$; $p = 0.07$). Nectar volume of flowers of the second day (daytime: 24-36 hours) had a similar pattern of flowers of the first day (as Fig. 3C), while the nectar concentration decreased slowly following the tendency observed in flowers of the first day (as Fig. 3D) (Guimarães *et al. under review*).

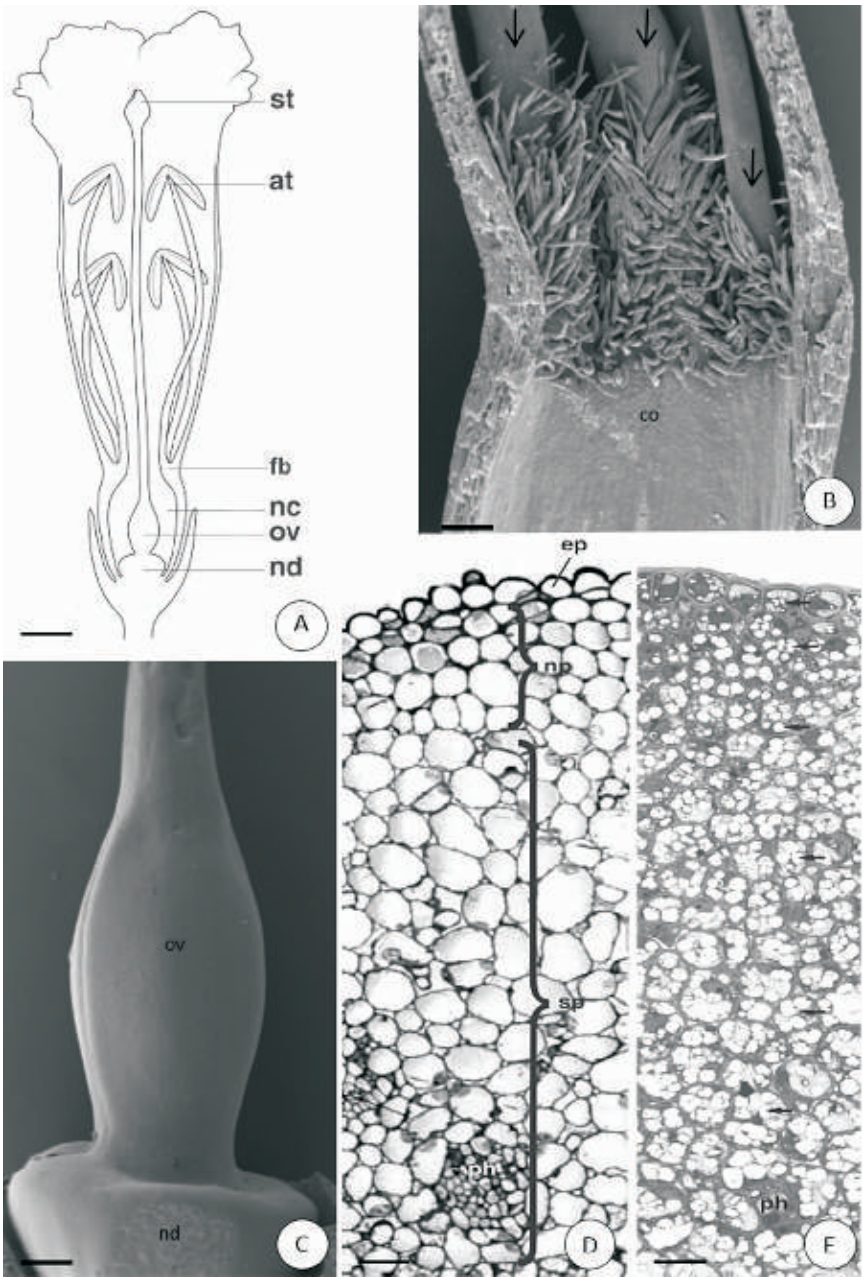


Fig. 2- Floral structure and nectary disk of *Anemopaegma album* (Bignoniaceae). **A.** A longitudinal section flower to highlight the arrangement of floral whorls, the placement of reproductive structures, the nectary and the nectar chamber (Drawn by Cesar Claro Trevelin). **B.** Longitudinal section through a floral tube showing the dilated base of filets covered by trichomes (arrows) observed with SEM. Note the basal portion of corolla tube completely glabrous comprising the nectar chamber. **C.** Lower portion of gynoecia showing the nectary disk below ovary observed under SEM. **D.** Cross section through a nectary showing the epidermis, nectary parenchyma and subnectary parenchyma supplied with phloem viewed with light microscopy **E.** Packed starch grains observed as white dots (arrows) through a cross section of the floral nectary; note their decrease in size and abundance from the subnectary parenchyma toward the epidermis. Phloem elements are shown in the subnectary parenchyma. Scale bar: (A) 0,6 cm, (B) 300 µm, (C) 250 µm, (D, E) 100 µm. (*Abbreviations*: at, anthers; co, corolla; ep, epidermis; fb, filet base; nc, nectar chamber; nd, nectary disc; np, nectary parenchyma; ov, ovary; ph, phloem; sp, subnectary parenchyma; st, stigma).

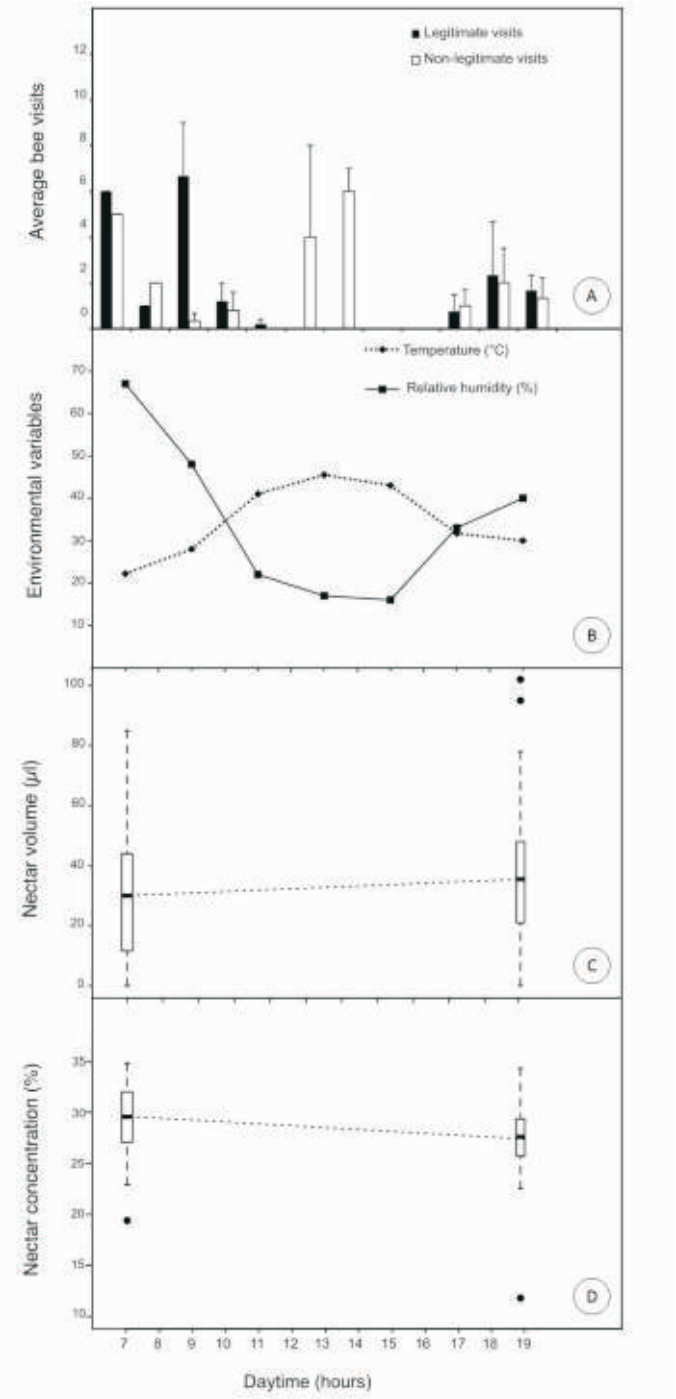


Fig. 3-Microclimate, bee visits and nectar descriptors in the study population of *Anemopaegma album* (Bignoniaceae) at the "State Park of Grão Mogol", MG, Brazil. **A.** The number of legitimate and non-legitimate visits performed by bees to *A. album* flowers in five days (replicates) during daylight hours. Data from 07:00 to 09:00 h correspond to just one replicate, so we have no information about standard deviation (bars). **B.** Air temperature and air relative humidity around the *A. album* plants, as measured one meter above the ground. **C-D.** Box plots showing the median of nectar volume (µl) and nectar concentration (%, w/w), respectively. The rectangles represent versions of the 1st and 3rd quartiles, the bars the min-max values not considering the dots that correspond to outliers (for details see function boxplot in R).

DISCUSSION

Anemopaegma album followed Gentry's (1974) expectation that species in the genus *Anemopaegma* would be pollinated by large- and medium-sized bees. This flower's architecture fits the size and behavior of these bee pollinators, allowing for nototribic pollen deposition and resulting in a pollination precision that could play a vital role in the successful sexual reproduction of *A. album* plants. In addition we verified an association between nectar secretion pattern and bee-pollinator visits. There was a high volume of nectar in the first period of the flower life span in accordance with pollinator visits frequency throughout the day, in contrast with the nectar robber visits, which was concentrated around midday.

The degree of *A. album* dependence on bees for sexual reproduction-Our experiments revealed that there was no seed production from selfing, indicating total dependence on pollination vectors, namely medium- and large-sized bees, for crossing and seed production. Our findings corroborate previous observations by Firetti-Leggieri et al. (2013) that considered *A. album* a diploid, self-incompatible species that exhibited no fruit or seed set in non-pollinated flowers.

Euglossini bees showed a visit behavior that favours outcrossing, since these bees visited few flowers per plant (1-3 flowers) and performed more sequences of visits per day than Bombini bees. Although Bombini bees visited twice more flowers than Euglossini bees per hour, these bees used to visit several flowers from the same plant in a sequence, probably promoting a gradual increase in geitonogamy from second flower on. This behavior may be at least partially responsible for the low fruit set in *A. album* which may be caused by pollen-stigma interference. Similar results concerning the low fruit set registered in *A. album* was described in several other Bignoniaceae species (Stephenson & Thomas 1977, Correia *et al.* 2006 Carvalho *et al*. 2007, Guimarães *et al.* 2008, Milet-Pinheiro & Schilindwein 2009), and it could also be related to pollen depletion by thieves. In our manual cross-pollination experiments, we isolated the flowers from visitors, preventing geitonogamy, and pollen was not a limiting factor. Thus, the recorded fruit set, 15 times higher than the natural fruit set, may be in accordance with both the hypothesis of pollen depletion by thieves and the hypothesis of pollen-stigma interference.

Floral structure and nectary features related to pollinators-Although the morphology of *A. album* flowers may reduce pollen-stigma interference because of approach and movement herkogamy (*sensu* Webb & Lloyd 1986) it works just for the pollen deposition on stigma from the same flower. In approach herkogamy, the pollen and stigmas are placed so that the stigmas are contacted first, as the visitor enters the blossom, and pollen is picked up subsequently or as the visitor departs (Webb & Lloyd 1986). In *A. album*, even if the pollinator touches the stigma when leaving the flower, the pollen may not be deposited on the receptive area of the stigma, which is adaxial positioned. In the meantime, if the floral visitor arrives with pollen load from flowers of the same plant, the pollen-stigma interference will occur anyway. Movement herkogamy is observed in *A. album* as stigmatic lobe closure after touching, and it also may prevent pollen-stigma interference (Milet-Pinheiro *et al.* 2009). Nevertheless, none of these mechanisms are efficient against geitonogamy, which can be avoided mainly by the pollinator behavior, what in fact reinforces the importance of Euglossini bees for sexual reproduction of *A. album*.

The nectary disk of *A. album* followed the pattern described by Rivera (2000) for *Anemopaegma*. The general anatomy of the floral nectary fits on the pattern proposed by Nepi (2007) and the starch grains stored in the secretory cells has been correlated to species that produces high volume of nectar in short periods (Pacini & Nepi 2007) as is the case of *A. album* flowers.

Relations among floral visitor activity, weather conditions and nectar availability-*Anemopaegma album* offered the highest volume of nectar, with higher concentration, at the moment of flower opening, as described for other Bignoniaceae species (Elias & Gelband 1975, Bertin 1982, Galetto *et al.* 1994), and follows the premise of Cruden *et al.* (1983), which consider that the beginning of secretion is adjusted to pollinator beginning activities. The period with high concentration of bee visits, including pollinators and nectar robbers occurred during the first hours of the day when the higher amount of nectar was available.

Weather may act as a constraint on pollination systems, and its effects can be direct, e.g., by affecting the hygrothermal balance of the pollinators, or it can also be indirect, e.g., through modifications in the floral reward (Corbet 1990). We postulate that the nectar storage at the bottom of the corolla tube in *A. album* minimizes the effects of environmental factors. This flower design reduces its desiccation through the maintenance of a microclimate inside the floral tube, especially inside the nectar chamber, which is closed in the apical region by the dilated base of the stamens covered by trichomes.

Our findings for the nectar volume and concentration in *A. album* flowers were not consistent with the results described for other bee-pollinated Bignoniaceae species by Galetto (2009). In general, large- and medium-sized bees have particular requirements in terms of nectar concentrations because they have high energy needs. These bees usually do not require water, especially because of substantial metabolic water production during flight (Nicolson 2009). Thus, large- and medium-sized bees are usually associated with flowers that produce small amounts of nectar that are more concentrated than they are in other insect-pollinated flowers (Baker & Baker 1983). By contrast, *A. album* flowers had a high volume of more diluted nectar, resulting in a considerable water source for bees, in addition to calories. So the diluted nectar of *A. album* flowers may be used by bees for rehydration, in addition to energetic purposes, similar to that described for some desert bees (see Nicolson 2009 for references). This diluted nectar offered by *A. album* flowers can be important for the mutualist bees, such as the pollinators, and also for the antagonist bees, such as the nectar robbers.

Nectar robbing (*sensu* Inouye 1980) was recorded in the pre-anthesis buds and functional flowers of *A. album*. The high frequency of floral larceny in Bignoniaceae has been related to the offer of suitable nectar, which is not accessible to bees with short mouthparts (Milet-Pinheiro & Schlindwein 2009). The effects of nectar robbers can be variable, for example, plants may be able to tolerate the effects of nectar robbing, producing enough nectar to supply all floral visitors with nectar resources (Irwin & Maloof 2002). In *A. album* plants, both pollinators and nectar robbers were observed visiting the flowers throughout the day. In addition, geitonogamy can be detrimental to a plant species, and it could become a problem for obligate outcrossing plant species, as in the case of *A. album* study population. In this situation, nectar robbers may have a positive effect on plant reproduction, reducing geitonogamy by changing pollinator foraging patterns (Maloof & Source 2000, Zhang *et al.* 2014).

Additionally, during the hottest and driest periods of the day, we observed just *Oxaea flavescens* visiting *A. album* flowers. This finding may be related to the yellow-brownish color of *O. flavescens*, which can provide lower heating rates when compared with the black and dark-green body color of the other bees that visit *A. album* flowers. In fact, Pereboom & Biesmeijer (2003) suggest that light-coloured large bees may have an advantage over large black bees in hot open habitats, such as Neotropical savannas, because they warm up less rapidly. Assuming that nectar secretion rate is constant throughout the day these bees could find a substantial volume of nectar that accumulated since 10:00 h, from when the frequency of visits decreased markedly. Thus, this bee species can take advantage of the hottest period of the day, when the other bees are not visiting the *A. album* flowers, to obtain nectar without competitors.

In our system, the antagonists are exploring the same resource that pollinators and both may perform, together, pressures on nectar traits and may affect the plant fitness of the study population. In fact, less than 50% of bee visits registered can result in pollen movement since around 53% of nectar collection visits was performed in an illegitimate way. This result suggests that the nectar is shared equally between the two functional groups of bees visiting the flowers of *A. album*.

Concluding remarks—*Anemopaegma album* showed obligate outcrossing and was dependent on bees for sexual reproduction, what may be favored especially by Euglossini bees visit behavior. Considering that it is a protandrous species, maximum nectar availability overlaps with the phase of greatest pollen presentation, improving male function and probably the pollination effectiveness, since the pollinator visits are also concentrated during this period.

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LITERATURE CITED

Baker HG & Baker I 1983 A brief historical review of the chemistry of floral nectar. In*:* Bentley B & Elias TE (eds.), *The biology of nectaries,* Pp. 126-152. Columbia University Press, New York, USA.

Bertin RI 1982 Floral biology, hummingbird pollination, and fruit production of trumpet creeper (*Campis radicans*, Bignoniaceae). *Am. J. Bot.* **69** 122-134.

Burkle LA, Irwin RE & Newman DA 2007 Predicting the effects of nectar robbing on plant reproduction: implications of pollen limitation and plant mating system. *Am. J. Bot.* **94** 1935-1943.

Carvalho AT, Santos-Andrade FG & Schlindwein C 2007 Baixo sucesso reprodutivo em *Anemopaegma laeve* (Bignoniaceae) no Parque Nacional do Catimbau, Pernambuco. *Revista Brasileira de Biociências* **5** 102-104.

Corbet SA 2003 Nectar sugar content: estimating standing crop and secretion rate in the field. *Apidologie* **34** 1–10.

Cruden RW, Hermann SM & Peterson S 1983 Patterns of nectar production and plat-pollinator coevolution. In: Bentley B & Elias TE (eds.), *The biology of nectaries,* Pp.80-125. Columbia University Press, New York, USA.

Elias TS & Gelband H 1975 Nectar: its production and functions in trumpet creeper. *Science* **189** 289-291.

Firetti-Leggieri F, Lohmann LG, Alcantara S, Costa IR & Semir J 2013 Polyploidy and polyembryony in *Anemopaegma* (Bignonieae, Bignoniaceae). *Plant Reproduction* **26** 43-53.

Galen C 1996 Rates of floral evolution: adaptation to bumblebee pollination in an alpine wildflower, *Polemonium viscosum*. *Evolution* **50** 120-125.

Galetto L 2009 Nectary and Nectar Features: Occurrence, Significance, and Trends in Bignoniaceae. *The Journal of Plant Reproductive Biology* **1**(2) 1-12.

Galetto L, Bernardello LM & Juliani HR 1994 Characteristics of secretion of nectar in *Pyrostegia venusta* (Ker-Gawl.) Miers (Bignoniaceae). *New Phytol.* **127** 465-471.

Gentry AH 1974 Coevolutionary patterns in Central American Bignoniaceae. *Annals of the Missouri Botanical Garden* **61** 728-759.

Giulietti AM, Pirani JR & Harley RM 1997 Espinhaço range region. Eastern Brazil. In: Davis SD, Heywood VH, Herrera-MacBryde O, Villa-Lobos J & Hamilton AC (eds.), *Centres of plant diversity. A guide and strategies for the conservation*, Vol. 3, The Americas, 397-404. WWF/IUCN, Cambridge, UK.

Guimarães E, Di Stasi LC & Maimoni-Rodella RCS 2008 Pollination biology of *Jacaranda oxyphylla* with an emphasis on staminode function. *Annals of Botany* **102** 699-711.

Herrera CM 2000 Flower-to-seedling consequences of different pollination regimes. *Ecology* **81**(1) 15-29.

Inouye DW 1980 The terminology of floral lacerny. *Ecology* **81** 1251-1253.

Irwin RE & Maloof JE 2002 Variation in nectar robbing over time, space, and species. *Oecologia* **133** 525–533.

Johnston MO 1991 Natural selection on floral traits in two species of Lobelia with different pollinators. *Evolution* **45** 1468-1479.

Kenoyer LA 1916 Environmental influences on nectar secretion. *Research Bulletin* **37** 219-232.

Lohmann LG 2003 *Phylogeny, classification, morphological diversification and biogeography of Bignonieae (Bignoniaceae).* Ph.D. thesis, University of Missouri-St. Louis, St.Louis.

Lohmann LG 2011 A new generic classification of Bignonieae (Bignoniaceae) based on molecular phylogenetic data and morphological synapomorphies. *Annals of Missouri Botanical Garden* (In Press).

Maloof JE 2001 The effects of a bumble bee nectar robber on plant reproductive success and pollinator behavior. *Am. J. Bot.* **88** 1960–1965.

Maloof EI & Source DW 2000 Are nectar robbers cheaters or mutualists? *Ecology* **81** 2651-2661.

Michener CD 2000 *The bees of the world.* The Johns Hopkins University Press, Baltimore, Maryland, USA.

Milet-Pinheiro P, Carvalho AT, Kevan PG & Schilindwein C 2009 Permanent stigma closure in Bignoniaceae: Mechanism and implications for fruit set in self-incompatible species. *Flora* **204** 82-88.

Milet-Pinheiro P & Schlindwein C 2009 Pollination in *Jacaranda rugosa* (Bignoniaceae): euglossine pollinators, nectar robbers and low fruit set. *Plant Biology* **11** 131-141.

Nepi M 2007 Nectary structure and ultrastructure. In: S. W. Nicolson SW, M. Nepi M & Pacini E (eds.), *Nectaries and nectar,* Pp. 129-166. Springer, Dordrecht, Netherlands.

Nicolson SW 1995 Direct demonstration of nectar reabsorption in the flowers of *Grevillea robusta* (Proteaceae). *Functional Ecology* **9** 584-588.

Nicolson SW 2009 Water homeostasis in bees, with the emphasis on sociality. *J. Exp. Biol.* **212** 429-434.

Nicolson SW & Nepi M 2005 Dilute nectar in dry atmospheres: nectar secretion patterns in *Aloe castanea* (Asphodelaceae). *International J. of Plant Science* **166** 227-233.

Pacini E & Nepi M 2007 Nectar production and presentation. In: Nicolson SW, Pacini E & Nepi M (eds.), *Nectaries and nectar*, Pp.167-214. Springer,Dordrecht, The Netherlands.

Petanidou T 2007 Ecological and evolutionary aspects of floral nectars in Mediterranean habitats. In: Nicolson SW, Pacini E & Nepi M (eds.), *Nectaries and nectar*, Pp. 343-376. Springer, Dordrecht, The Netherlands.

R Core Team 2014 R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL http://www.R-project.org/.

Ramos AM, Santos LAR & Fortes LTG 2009 Normais climatológicas do Brasil. Instituto Nacional de Meteorologia, Brasília, DF, Brazil. Pp. 465.

Rapini A, Ribeiro PL, Lambert S & Pirani JR 2008 A flora dos campos rupestres da Cadeia do Espinhaço. *Megadiversidade* **4** 16-24.

Rivera GL 2000 Nuptial nectary strucuture of Bignoniaceae from Argentina. *Darwiniana* **38** 227-239.

Silva EM & Dean BB 2004 Patterns of floral nectar production of onion (*Allium cepa* L.) and the effects of environmental conditions. *J. Am. Soc. Hort. Sci.* **129** 299-302.

Stephenson AG & Thomas WW 1977 Diurnal and nocturnal pollination of *Catalpa speciosa* (Bignoniaceae). *Systematic Botany* **2** 191-198.

Sun SG, Huang SQ & Guo YH 2013 Pollinator shift to managed honeybees enhances reproductive output in a bumblebee-pollinated plant. *Plant Systematics and Evolution* **299** 139-150.

Walter H 1986 Vegetação e Zonas Climáticas. Editora Pedagógica e Universitária. São Paulo, São Paulo, Brazil. Pp. 325.

Webb CJ & Lloyd DG 1986 The avoidance of interference between the presentation of pollen and stigmas in angiosperms II. Herkogamy. *New Zealand J. of Botany* **24** 163-178.

Zhang YW, Zhao JM & Inouye DW 2014 Nectar thieves influence reproductive fitness by altering behavior of nectar robbers and legitimate pollinators in *Corydalis ambigua* (Fumariaceae). *J. of Ecology* **102** 229–237.